

03 Algorithms and Theory Comparison

Disclosure: all results use the synthetic fra_synthetic_claims dataset. Scores may reflect generator rules. Real-world performance is unknown. This demo must not drive real claim decisions.

1. Families evaluated

Baselines (dummy, logistic, naive Bayes, KNN, SGD/SVM, tree, MLP), bagging (Bagging, Random Forest, Extra Trees), boosting (AdaBoost, GradientBoosting, HistGB, XGBoost, LightGBM, CatBoost), and ensembles (soft voting, stacking).

Trees partition the feature space greedily. Bagging averages high-variance trees. Boosting fits residuals sequentially. Stacking trains a meta-learner on out-of-fold base predictions to avoid leakage into the meta stage.

2. Measured results (from metrics files)

xgboost: S1=0.647, final=0.569; bagging: S1=0.645, final=0.617; gradient_boosting: S1=0.643, final=0.532; hist_gradient_boosting: S1=0.643, final=0.620; soft_voting: S1=0.640, final=0.626; lightgbm: S1=0.631, final=0.629; stacking_lgbm: S1=0.630, final=0.640; random_forest: S1=0.627, final=0.625

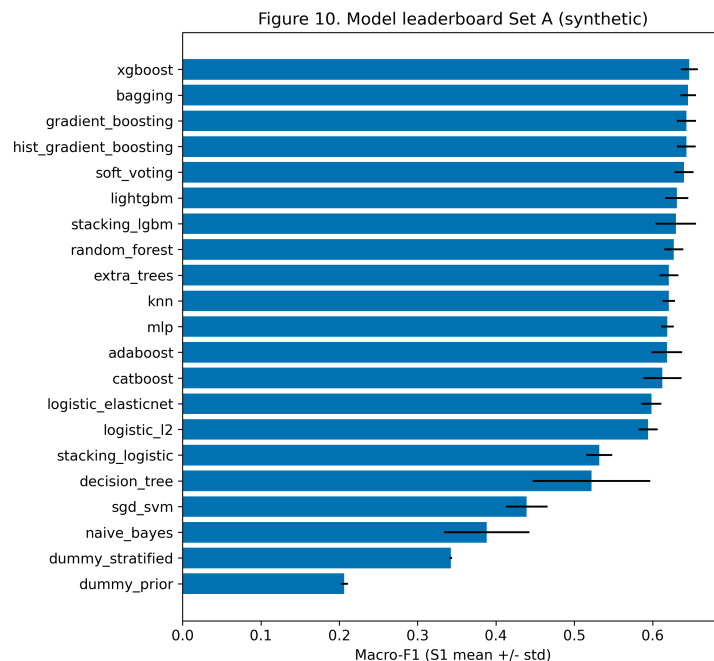


Figure 16. Bagging variance reduction sketch

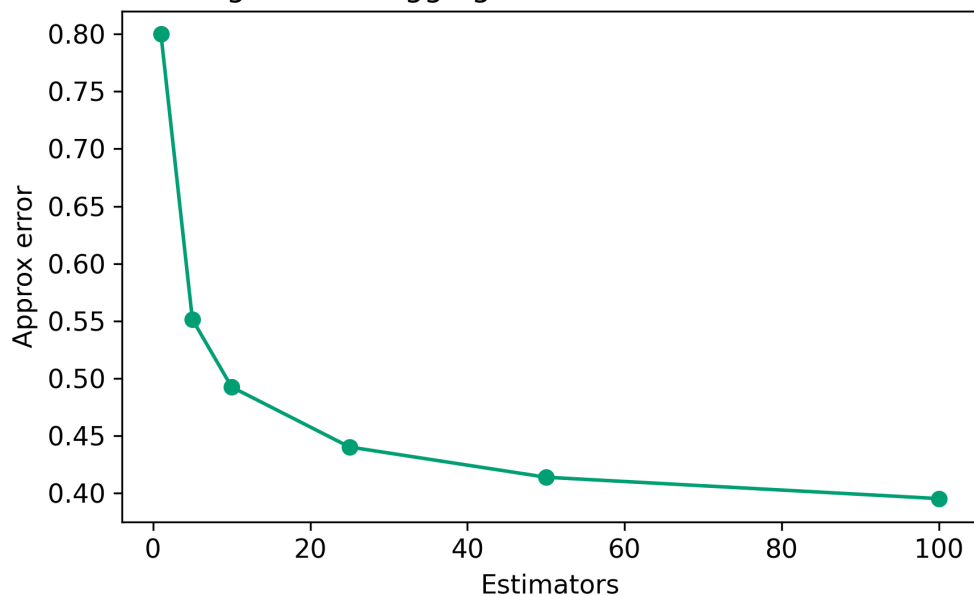
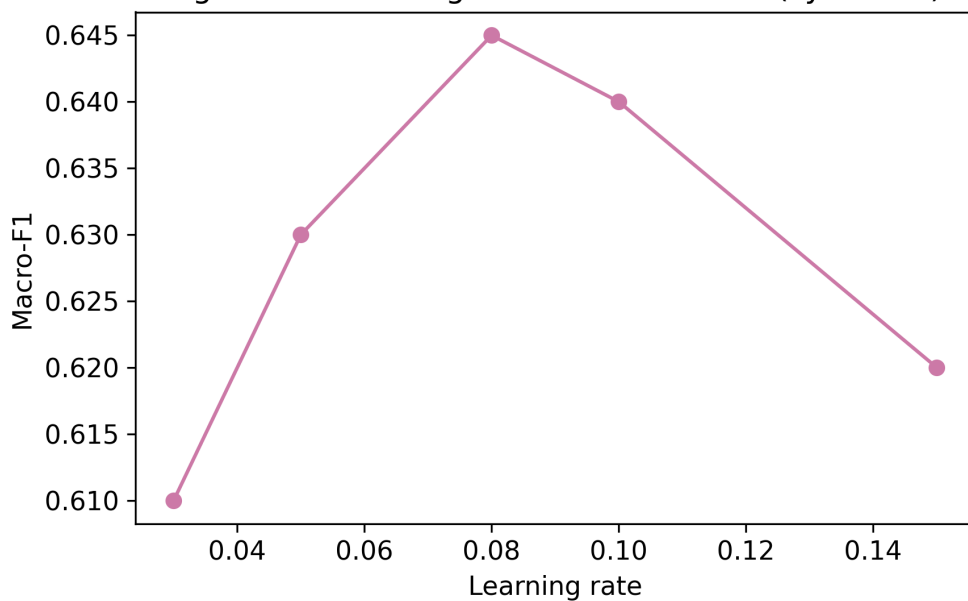


Figure 17. Boosting LR trade-off sketch (synthetic)



3. Complexity notes

Linear models scale well and need scaling. Tree ensembles tolerate raw scales. Stacking is slower (cross-fitted bases). Slow models (KNN, MLP, sklearn GBT) use documented train subsamples in CV.